

CANDIDATE BUILD CHALLENGE BRIEF

Energy Systems Graduate Recruitment 2026 · SuperCharge SG × SuperAgent.ai

Three Challenges · Fourteen Days · Build What Matters

Challenge 1	Challenge 2	Challenge 3	Total Score
Telegram Bot (base) +Bonus: WhatsApp	Smart Energy Dashboard	AGV Brain Proposal (Mandatory)	100 pts + 20 bonus
35 pts + 20 WhatsApp bonus	35 pts	30 pts	Deadline: 14 days from issue

Welcome to the SuperCharge SG build challenge. You have been shortlisted as a Priority candidate based on your application. This brief outlines your three build challenges — two software products and one hardware design proposal. All three are real products that SuperCharge SG intends to build and deploy. Your submission will be evaluated on what actually works, not what you describe. Build it. Ship it. Show us.

14 days

You have 14 days from the date this brief is issued. Deadline is 11:59 PM SGT, 25th March 2026 (Day 14). Bonus points awarded for early submission (see scoring).

3 challenges

All 3 must be attempted. Challenge 3 (AGV Brain) requires a proposal document, not code.

WhatsApp bonus

Build on WhatsApp Business API instead of (or in addition to) Telegram and earn +20 bonus points.

Early submission bonus

Submit before Day 10 and earn +5 pts. Submit before Day 7 and earn +10 pts.

Live demo required

Challenges 1 and 2 must be accessible via a live URL or bot link. No screenshots accepted.

Knowledge base

You have been given a companion Knowledge Base PDF. Use it. It contains tips, tech stacks, and context.

1

AI Customer Support Chatbot

Telegram (base) · WhatsApp Business API (+20 bonus) · RAG-powered · Live deployed

What You Are Building

SuperCharge SG handles dozens of customer enquiries every week across EV charging and solar installation: pricing, TR25 compliance, OCPP errors, ECIS export credits, installation timelines, and maintenance requests. You are building an AI-powered customer support bot that handles these conversations automatically — grounded in real SuperCharge knowledge, not generic AI guesswork. The bot must be live and reachable during evaluation. A link or bot username must be submitted.

Platform Choice

TELEGRAM (Base)

Build using python-telegram-bot or Telegraf.js. Easier to set up — no business account required. Earns base 35 points.

WHATSAPP BUSINESS API (+20 bonus pts)

Build using Twilio WhatsApp sandbox OR WhatsApp Cloud API (Meta). Harder to set up but earns +20 bonus points because WhatsApp is SuperCharge SG's primary customer channel. Worth the effort.

What the Bot Must Do

- ✓ Answer customer questions about SuperCharge SG services — EV chargers, solar panels, pricing, TR25, ECIS credits, installation process — using a RAG knowledge base. No hallucination. Answers must be sourced from your KB.
- ✓ Capture leads: when a user expresses interest, collect name, email, and enquiry type and save to a Google Sheet or Airtable.
- ✓ Escalate intelligently: when the bot cannot answer with confidence, send a human handoff alert (email or Slack) and tell the user a human will follow up.
- ✓ Maintain conversation context: a follow-up question like 'What about the warranty?' must understand it refers to the previous topic.
- ✓ Handle at least these intents: EV Charger enquiry, Solar Panel enquiry, Price / Quote request, Maintenance / Fault report, General FAQ, Human agent request.

Deliverables

- ✓ Live bot link (Telegram username or WhatsApp number) — must be active during the evaluation window
- ✓ GitHub repository with full source code (public or shared with recruiter)
- ✓ RAG knowledge base files (source documents, chunking strategy documented in README)
- ✓ Lead capture destination: share view access to the Google Sheet / Airtable where leads are stored
- ✓ Escalation proof: screenshot or log showing human handoff alert triggered correctly
- ✓ README: architecture diagram, RAG pipeline explanation, deployment steps, known limitations
- ✓ 3-minute Loom video: walk through a live conversation covering at least 3 different intents

Scoring

Criterion	Points	How to earn full marks
Live deployment	10	Bot is reachable via link during evaluation. Responds within 5 seconds. No crashes.
RAG accuracy — no hallucination	10	Every factual answer cites a real SuperCharge SG fact. Test: 5 curveball questions asked by evaluator.
Intent coverage	5	All 6 required intents handled correctly in the evaluation conversation.
Lead capture flow	5	Lead saved to Sheet/Airtable correctly. Confirmation sent to user.
Escalation logic	5	Human handoff triggers when evaluator says 'speak to a person' or bot hits low confidence.
WhatsApp BONUS	+20	Bot deployed on WhatsApp Business API and fully functional.
Early submission (Day 7)	+10	All 3 challenges submitted and functional by end of Day 7.
Early submission (Day 10)	+5	All 3 challenges submitted and functional by end of Day 10.

TIP: The Knowledge Base PDF contains a full SuperCharge SG product and FAQ reference, suggested RAG chunking strategies, code architecture tips, and WhatsApp Cloud API setup guides. Read it before you start.

2

Smart Energy Dashboard SaaS

Real-time solar + EV monitoring · Multi-site · AI anomaly detection · Auto PDF report

What You Are Building

SuperCharge SG's enterprise clients — condominiums, commercial buildings, industrial parks — have both solar installations and EV charging networks, but no unified view of their energy ecosystem. You are building the single pane of glass that combines both: live solar generation, EV charging activity, grid export credits (ECIS), CO2 savings, and an AI layer that detects anomalies and sends weekly plain-English performance digests. This must be a working web application — live, multi-tenant, with real data pipelines and a real ML model.

Core Requirements

- ✓ Real-time data pipeline: simulate solar inverter output (power kW, cumulative kWh, temperature, irradiance) and EV charging sessions (kWh delivered, session duration, revenue SGD) for at least 3 virtual sites. Data must update live in the dashboard — not static mock data.
- ✓ ML anomaly detection: train and deploy an anomaly detection model (Isolation Forest, LSTM, or Prophet) that flags when a solar site's actual yield deviates more than 15% below expected yield (corrected for irradiance). Use NASA POWER API for real Singapore irradiance data.
- ✓ Multi-tenant isolation: create at least 2 client accounts. Client A must not see Client B's data. JWT-based auth with org-scoped queries.
- ✓ AI weekly digest: generate a plain-English weekly energy summary per client using an LLM — auto-sent by email.
- ✓ Auto monthly PDF report: branded, downloadable — client name, site summary, savings chart, anomaly log, ECIS credits earned.
- ✓ ECIS credit tracker: calculate and display ECIS export credits based on Singapore's current exported unit rate (SGD 0.218/kWh baseline — verify in Knowledge Base).

Deliverables

- ✓ Live URL — dashboard accessible during evaluation window (Vercel, Railway, or Docker Compose)
- ✓ Two test client login credentials provided to evaluator
- ✓ GitHub repository with full source code
- ✓ Working anomaly detection: evaluator will inject a fault into the simulated data and verify it is flagged
- ✓ Sample AI weekly digest email — forward a generated digest to support@supercharge.sg
- ✓ Sample monthly PDF report — attach one generated report to your submission
- ✓ README: data model, anomaly detection methodology, multi-tenant design, NASA API integration, deployment guide
- ✓ 3-minute Loom: demo multi-tenant login, live data, anomaly trigger, PDF download

Scoring

Criterion	Points	How to earn full marks
Live dashboard — real-time data	8	Data updates live without page refresh. At least 3 virtual sites visible.

ML anomaly detection	9	Model correctly flags injected fault. False positive rate discussed in README.
Multi-tenant isolation	7	Client A cannot access Client B data. Verified by evaluator with both logins.
AI weekly digest quality	5	LLM-generated digest is accurate, non-generic, and client-ready.
Monthly PDF report	4	PDF is branded, correctly calculated, and downloadable on demand.
ECIS credit accuracy	2	ECIS credit calculation matches Singapore's published export rate.

TIP: The Knowledge Base PDF contains SuperCharge SG's full solar and EV product specs, Singapore ECIS scheme details, NASA POWER API integration guide, and recommended data model for multi-tenant time-series storage.

3

AGV Grass Cutter Brain — Design Proposal

MANDATORY FOR ALL · Hardware design · BOM · Wiring diagram · No physical build required

IMPORTANT: This is a DESIGN PROPOSAL challenge. You do not build or solder anything. You produce a complete engineering proposal that SuperCharge SG's team will use to procure components and build the system. Think of yourself as the engineer briefing the build team. Your proposal must be detailed enough that a competent hardware engineer can follow it without asking you any clarifying questions.

What You Are Designing

SuperCharge SG is building an autonomous grass cutter for grounds maintenance at solar farm sites and commercial properties. The target is a universal plug-and-play AGV Brain module that wires in-line between an industrial PWM radio receiver (TADA B1900 class: 24V DC input, PWM outputs for engine start/stop, electric push rods, throttle, damper, signal outputs 1 and 2) and the grass cutter's drive systems. When the operator flicks a switch, control passes from the RC transmitter to the onboard compute module. When they flick it back, the RC transmitter resumes full control instantly.

TADA B1900 Receiver — Signal Reference

Terminal	Signal Type	AGV Brain Must	Safety Priority
Engine Start (+/-)	Digital relay	Read in MANUAL. Replicate with GPIO relay in AUTO.	HIGH
Engine Stop (+/-)	Digital relay	ALWAYS honour in BOTH modes — never override E-stop.	SAFETY-CRITICAL
DC Push Rod 1 (+/-)	PWM motor drive	PWM capture in MANUAL. Override with motor driver in AUTO.	MEDIUM
DC Push Rod 2 (+/-)	PWM motor drive	Same as Push Rod 1.	MEDIUM
DC 24V Input (+/-)	Power supply	Step down to 5V/12V for compute via buck converter.	HIGH
Signal Output 1 (JST)	RC PWM servo	Capture duty cycle in MANUAL. Override in AUTO.	HIGH
Signal Output 2 (JST)	RC PWM servo	Same as Signal Output 1.	MEDIUM
Throttle (+/-)	PWM/analog	Capture and override for speed control in AUTO.	HIGH
Damper (+/-)	PWM/analog	Capture and override for blade engagement.	MEDIUM
ANT (SMA)	RF antenna	Do NOT modify — pass through untouched.	N/A

What Your Proposal Must Cover

- ✓ System block diagram: RC Transmitter > TADA B1900 Receiver > AGV Brain (with MANUAL/AUTO switch) > Grass Cutter actuators. Drawn clearly with signal flow arrows.
- ✓ Compute module selection: compare and justify your choice among Raspberry Pi CM4, Jetson Nano, or RPi 4B + STM32. One must handle real-time PWM; the other handles navigation.
- ✓ PWM intercept circuit: describe how your module reads the existing PWM signals and either passes them through (MANUAL) or replaces them (AUTO) without damaging the RC receiver.

- ✓ **MANUAL / AUTO mode switching:** hardware switch wiring and software state machine. What happens on signal loss? Failsafe must default to MANUAL.
- ✓ **Navigation stack:** propose ROS2 Nav2 (with GPS waypoints) or ArduPilot Rover. Justify for an outdoor grass-cutting scenario. What sensors are needed?
- ✓ **Safety interlocks:** blade stop on tilt, bump sensor E-stop, operator presence detection, geofence boundary enforcement.
- ✓ **Power management:** 24V system. Size a buck converter for compute power. Add supercapacitor or small LiPo for graceful shutdown on power loss.
- ✓ **Full Bill of Materials:** every component, specification, supplier, unit cost in SGD, quantity, total. Target: under SGD 900.
- ✓ **Integration wiring guide:** step-by-step instructions for connecting the AGV Brain to a TADA B1900 class receiver with pin mapping.
- ✓ **Risk register:** top 5 technical risks with likelihood, impact, and mitigation strategy.

Deliverables

- ✓ PDF proposal document — minimum 8 pages, professional formatting
- ✓ System block diagram (draw.io, Visio, Lucidchart, or clearly hand-drawn and scanned)
- ✓ Bill of Materials — Excel or Google Sheets with SGD pricing (share link or attach)
- ✓ TADA B1900 signal intercept wiring diagram with pin/terminal mapping
- ✓ 500-word executive summary: why your design is universal and plug-and-play to any off-the-shelf grass cutter using a comparable PWM RC receiver

Scoring

Criterion	Points	How to earn full marks
System architecture clarity	6	Block diagram is clear, signal flow is correct, all TADA B1900 terminals addressed.
PWM intercept design soundness	6	Intercept method is technically valid and non-destructive to the RC receiver.
Navigation stack appropriateness	5	Stack choice (ROS2 / ArduPilot) is justified for outdoor autonomous grass cutting.
Safety interlock completeness	5	All critical safety scenarios addressed — tilt, bump, geofence, failsafe-to-manual.
BOM accuracy & SGD costing	5	BOM is complete, component specs are real, costs sourced from real suppliers. Under SGD 900.
Integration guide practicality	3	A technician could wire this up following your guide without additional questions.

TIP: The Knowledge Base PDF has a full TADA B1900 signal reference table, compute module comparison, ROS2 vs ArduPilot decision guide, PWM intercept MUX circuit design, and a sample BOM template with Singapore-market supplier references.

Submission Checklist & Timeline

Day 0	Days 1–2	Days 3–5	Days 6–8	Days 9–11	Days 12–13	Day 14
Receive brief. Read Knowledge Base. Set up repos and bot accounts.	Architecture planning. Bot account setup. Dashboard scaffolding. AGV proposal outline.	Core bot logic + RAG. Dashboard data pipeline + ML model. AGV BOM research.	Bot intents + lead capture + escalation. Dashboard multi-tenant + AI digest. AGV wiring diagram.	Testing all three. Fix critical bugs. Draft READMEs and proposal docs.	Record Loom videos. Final polish. AGV proposal PDF finalised.	Submit all links and files by 11:59 PM SGT. Email to: support@supercharge.sg

EARLY SUBMISSION BONUS: Submit all 3 challenges fully completed before Day 7 (11:59 PM SGT) = +10 bonus points. Before Day 10 = +5 bonus points. Partial submissions do not qualify — all 3 challenges must be live and submitted.

Master Submission Checklist

CHALLENGE 1 — CHATBOT	CHALLENGE 2 — DASHBOARD	CHALLENGE 3 — AGV BRAIN
✓ Live Telegram bot link ✓ WhatsApp bot number [BONUS] ✓ GitHub repo URL ✓ Google Sheet / Airtable link ✓ Escalation screenshot ✓ README + architecture diagram ✓ 3-min Loom video	✓ Live dashboard URL ✓ 2x client login credentials ✓ GitHub repo URL ✓ AI weekly digest (forwarded email) ✓ Monthly PDF report (attached) ✓ README + deployment guide ✓ 3-min Loom video	✓ PDF proposal (min 8 pages) ✓ System block diagram ✓ BOM — Excel / Google Sheets ✓ TADA B1900 wiring diagram ✓ 500-word executive summary ✓ Risk register (top 5 risks)

Challenge	Max Points	Bonus Points	Total Possible
1 — Chatbot (Telegram)	35	+20 (WhatsApp)	55
2 — Smart Energy Dashboard	35	—	35
3 — AGV Brain (Mandatory)	30	—	30
Early Submission	—	+10 (Day 7) / +5 (Day 10)	+10
TOTAL		+30 max bonus	130

Good luck. Build something you would be proud to demo live.
Submissions to: support@supercharge.sg — SuperCharge SG Recruitment Team